mechanics, at least classically. The earliest application of quantum mechanics was to the motion of nonrelativistic particles.

I showed earlier (in Lectures 4 and 8) that Poisson brackets play the same mathematical role in classical mechanics as commutators do in quantum mechanics. Written with these constructs, the classical and quantum mechanical equations of motion are almost identical in form. In particular, the Hamiltonian comes into play in the same way with Poisson brackets as it does with commutators. So, if you want to write down the quantum mechanical equations of a system whose classical physics you already know, it's very reasonable to try using the classical Hamiltonian, translated into operator form.

For a nonrelativistic free particle, the natural Hamiltonian to try is $p^2/2m$. When we say the particle is free, what we really mean is that no forces are acting on it, and therefore we can ignore potential energy. All we care about is the kinetic energy, which is defined as

$$T = \frac{1}{2}mv^2.$$

As you recall, the momentum for a classical particle is

$$p = mv.$$

The Hamiltonian is just the kinetic energy, which we can write in terms of the momentum p. This gives us

$$H = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

for the Hamiltonian of a classical nonrelativistic free particle. Unlike the right-going zaxon of the previous example, the energy of this particle does not depend on its direction of motion. That's because the energy is proportional to p^2 rather than p itself. So we'll start with a particle whose